

THE PHYSICAL NATURE OF CATALYTIC ACTIVITY DUE TO THE MOLECULAR ENVIRONMENT IN TERMS OF INTERMOLECULAR INTERACTION THEORY: DERIVATION OF SIMPLIFIED MODELS*

W. ANDRZEJ SOKALSKI

Institute of Organic and Physical Chemistry I-4, Wyb. Wyspiańskiego 27, 50-370 Wrocław (Poland)

(Received April 3, 1984; accepted June 4, 1984)

Summary

The catalytic or inhibitory activity of a molecular environment has been quantitatively represented as the difference between the interaction energies of the transition complex and the substrate complex with catalyst, and it has been further decomposed into the electrostatic multipole, electrostatic penetration, exchange, induction and dispersion contributions defined within exchange perturbation theory. A sample analysis of catalytic activity of the first hydration shell has been performed for the reaction $\text{CO}_2 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{CO}_3$ within the LCAO MO SCF *ab initio* approach. Results indicate a significant contribution of the electrostatic multipole term in the overall catalytic activity. However, the remaining electrostatic penetration, exchange, induction and dispersion interactions seem to be non-negligible but cancelling each other to a great extent, justifying some intuitively assumed electrostatic hypotheses on the nature of catalytic activity. More approximate models of gradually increasing simplicity have been briefly outlined based on cumulative atomic multipole expansion and nonempirical short range atom-atom potentials, making it possible to deal with much larger molecular systems and to predict the sites of the most effective solvation or the charge distribution of the optimal catalyst for a given reaction.

Introduction

Predicting the catalytic or inhibitory activity** of any molecular environment seems to be prohibitively complex and expensive for most chemical reactions. To reduce the large size of the reactant + environment supersystem, usually only the reactants are explicitly included in quantum

*Invited lecture at the Sanibel International Symposium on Quantum Biology and Quantum Pharmacology, Palm Coast, FL, U.S.A., 1983.

**Although the term 'catalytic' has more general meaning, for brevity we will use it in a more technical sense to describe the microenvironmental effects (inhibitory or catalytic) on chemical reactions expressed in the form of activation barrier lowering Δ .

chemical calculations and the remaining environment is handled in less rigorous fashion. Several electrostatic models of varying sophistication [1 - 9] are often employed in simulating the environment (solvent, enzyme active site *etc.*), but there is still not enough evidence that other kinds of interactions are indeed negligible. In addition methods based on macroscopic models cannot properly describe the anisotropic environment. Results obtained on the semiempirical level cannot be conclusive enough, as they could depend strongly on the choice of parameters. Besides, methods employing the intermolecular ZDO (Zero Differential Overlap) approximation cannot correctly reproduce exchange or charge transfer interactions [10]. Finally, the representation of the molecular charge distribution in the form of molecular dipoles or atomic charges may not be accurate enough to represent electrostatic multipole components [11] or molecular potentials [12].

The main aim of this work is to analyse the nature of catalytic or inhibitory activity in any molecular environment on the nonempirical level by expressing it in terms *related* to well-defined components of intermolecular forces. To our knowledge this is the first attempt to describe the relative importance of various contributions of such activity. Information about the nature of catalytic activity of molecular surrounding will be helpful in the systematic derivation of more approximate models, which are briefly outlined at the end of this work. This approach leads also to a simpler theoretical model based on differences of molecular electrostatic potentials, allowing the design of synthetic catalysts in a less empirical and more organized way [13].

In the present study the nature of catalytic activity due to the first solvation shell has been analyzed for the CO₂ hydration reaction within the 4-31G basis set employing a SCF interaction energy decomposition scheme corrected for basis set superposition error (BSSE) [14].

Catalytic activity contributions

The catalytic activity of the molecular surrounding can be quantitatively expressed as the activation barrier height (B) lowering Δ [7, 13, 15] which is related directly to the ratio of reaction rate constants; k_G in the gas phase and k_C in the presence of catalyst, if only entropy and vibrational contributions are neglected.

$$RT \ln(k_G/k_C) = \Delta H_C^0 - \Delta H_G^0 - T(\Delta S_C^0 - \Delta S_G^0) \cong B_C - B_G \equiv \Delta \quad (1)$$

Although the entropy term could sometimes be non-negligible, especially for weakly interacting systems [16], recent studies by Williams *et al.* [17] indicate that the enthalpy contribution remains the major source of catalytic activity. Negative Δ values indicate catalytic and positive inhibitory activity of any molecular environment. The time-consuming calculations of ac-

tivation barriers are very costly and results seem to be strongly basis set dependent [15].

Fortunately, the activation barrier lowering Δ may be alternatively written [7, 13, 15] as the difference of interaction energies of catalyst C with transition complex TC and substrate complex SC

$$\Delta \equiv B_C - B_G = \Delta E_{TC, C} - \Delta E_{SC, C} \quad (2)$$

This way, the problem of expressing the catalytic activity can be formulated in terms of theory of intermolecular forces.

The most complete theory of intermolecular interactions seems to be the exchange perturbation approach. Its numerous variants proposed since 1965 have been recently generalized by Jeziorski and Kołos [18]. The total *two-body* interaction energy ΔE is there decomposed into a sum of additive components, of which the most significant are the first-order electrostatic $E_{EL}^{(1)}$, exchange $E_{EX}^{(1)}$ terms and the second-order induction $E_{IND}^{(2)}$, dispersion $E_{DISP}^{(2)}$ contributions.

$$\Delta E = E_{EL}^{(1)} + E_{EX}^{(1)} + E_{IND}^{(2)} + E_{DISP}^{(2)} + \dots \quad (3)$$

Practical use of exchange perturbation theories seems to be limited to rather weak interactions and small systems because of the time-consuming integral transformations involved. In 1976 Jeziorski and van Hemert [19] demonstrated that the first three terms in the above expression (3) are approximately equivalent to the SCF variational interaction energy corrected for nonphysical basis set superposition error (BSSE) [20] by means of the Boys counterpoise method [21]. Moreover, variational SCF approach may be used for more strongly interacting systems and the use of costly integral transformation is avoided. The SCF interaction energy, corrected for BSSE, ΔE_{SCF} can be also decomposed into electrostatic $E_{EL}^{(1)}$, exchange $E_{EX}^{(1)}$ and higher order induction $E_{IND}^{(R)}$ contributions arising from locally excited configurations $E_{IND, LE}^{(R)}$ or charge transfer configurations $E_{IND, CT}^{(R)}$ [14].

$$\Delta E_{SCF} = E_{AB} - E_{A(B)} - E_{B(A)} = E_{EL}^{(1)} + E_{EX}^{(1)} + E_{IND, LE}^{(R)} + E_{IND, CT}^{(R)} \quad (4)$$

The physical meaning of the above-mentioned contributions is as follows. The first-order electrostatic term $E_{EL}^{(1)}$

$$E_{EL}^{(1)} = \langle \psi_0 | \hat{V} | \psi_0 \rangle \quad (5)$$

represents the coulomb interactions of the static charge distributions of A and B subsystems described by the product of their single determinant ground state wavefunctions $\psi_0 = \phi_0^A \phi_0^B$. The $\hat{V}$ operator refers to the interactions between all electrons and nuclei of A and B. This term, determined mainly by the magnitude and orientation of multipole moments, is the most anisotropic and significant contribution to overall interaction energy, especially for polar or charged systems. Electrostatic interaction energy $E_{EL}^{(1)}$ can be further decomposed:

$$E_{EL}^{(1)} = E_{EL, MTP}^{(1)} + E_{EL, PEN}^{(1)} \quad (6)$$

The short range penetration component due to mutual overlap of both interacting charge distributions $E_{\text{EL, PEN}}^{(1)}$ vanishes exponentially with increasing intermolecular separation R , in contrast to the long-range multipole component $E_{\text{EL, MTP}}^{(1)}$ easily evaluated from expansion:

$$E_{\text{EL, MTP}}^{(1)} = \sum_{a \in A} \sum_{b \in B} \sum_{l_a, l_b=0}^{\infty} C(l_a, l_b) R_{ab}^{-(l_a+l_b+1)} \quad (7)$$

where R_{ab} denotes interatomic distance and $C(l_a, l_b)$ is composed from the permanent atomic multipole moments [12]. The use of the cumulative atomic multipole moments (CAMP) instead of traditionally applied molecular multipole moments (MMM) results in much better convergency of the multipole expansion (7), especially for shorter intermolecular separations.

The remaining purely quantum mechanical first-order exchange term $E_{\text{EX}}^{(1)}$ originates from the Pauli exclusion principle and can be generally written

$$E_{\text{EX}}^{(1)} = \langle \psi_0 | \hat{\mathcal{A}} \hat{V} | \psi_0 \rangle / \langle \psi_0 | \hat{\mathcal{A}} | \psi_0 \rangle - E_{\text{EL}}^{(1)} \quad (8)$$

where $\hat{\mathcal{A}}$ denotes operator (antisymmetrizer) corresponding to the intermolecular exchange of electrons (due to their indistinguishable character as fermions). This term is always repulsive for closed shell systems and vanishes exponentially with distance.

The second-order interaction energy $E^{(2)}$ can be expressed as

$$E^{(2)} = \sum_{m=0}^{\infty} \langle \psi_0 | \hat{V} - \langle \psi_0 | \hat{V} | \psi_0 \rangle | \psi_m \rangle^2 / (E_0 - E_m) \quad (9)$$

and is expanded in terms of the complete set of excited configurations ψ_m , where E_0 and E_m denote the energies of noninteracting supermolecules in their ground and m th excited state. The contribution arising from the interaction of all locally excited (LE) states in only one molecule at a time, $\phi_0^A \phi_m^B$ or $\phi_m^A \phi_0^B$, is the classical induction energy $E_{\text{IND, LE}}^{(2)}$ whereas the remainder of the induction term can be ascribed to the excited charge transfer configurations $\phi^{A-} \phi^{B+}$ or $\phi^{A+} \phi^{B-}$. Both induction energy contributions $E_{\text{IND, LE}}^{(2)}$ and $E_{\text{IND, CT}}^{(2)}$ are always attractive and the usually insignificant $E_{\text{IND, LE}}^{(2)}$ term varies approximately as $\sim R^{-4}$, whereas the short-range $E_{\text{IND, CT}}^{(2)}$ has exponential character. Finally, the long-range second-order dispersion term (always attractive), which varies with distance approximately as $\sim R^{-6}$, represents the contribution from states excited simultaneously in both interacting molecules $\phi_m^A \phi_n^B$. In general, the magnitude of the second-order interactions is related to polarizabilities of interacting molecules. $E_{\text{IND, CT}}^{(2)}$, as defined within exchange perturbation theory, is not additive to the remaining $E_{\text{IND, LE}}^{(2)}$ contribution because locally excited configurations and charge transfer configurations form an overcomplete set [10, 14]. Fortunately, in the variational formulation used in this study [14] both terms are additive; this is an additional advantage of our approach. Moreover, in

contrast to the standard SCF interaction energy decomposition scheme [22], in our approach the nonphysical and frequently significant BSSE has been removed from the interaction energy components.

The most serious problem encountered in practical calculations on the nonempirical level is the quite strong basis set dependence of the results. This dependence may be reduced to some extent by removing *intermolecular* BSSE [11, 20, 23] resulting from the extension of the monomer basis set to the dimer basis set size. The remaining dependence of the corrected ΔE_{SCF} interaction energy originates mainly from the multipole electrostatic term $E_{\text{EL, MTP}}^{(1)}$ and could be ascribed to uncorrected *intramolecular* BSSE [11] originating from the extension of the atomic basis set to the size of the monomer basis set. Namely, the lack of the basis set balance within isolated molecules is responsible for nonphysical polarization of the molecular charge distribution. This effect may be substantially reduced by choosing well-balanced atomic basis sets [20, 24] or applying appropriate procedures [25, 26]. Reasonable results can be obtained without employing extended basis sets [11], except for the dispersion term $E_{\text{DISP}}^{(2)}$ not included in the SCF approach. In the latter case, the use of extended basis sets with polarization functions appears to be essential [27]. For large molecular systems where the use of extended basis sets will be prohibitively expensive, we estimate $E_{\text{DISP}}^{(2)}$ from atom-atom C_6 coefficients [11], fitting accurate dispersion energies available from the literature.

$$E_{\text{DISP}}^{(2)} = - \sum_{a \in A} \sum_{b \in B} (C_6^{aa} C_6^{bb})^{1/2} R_{ab}^{-6} \quad (10)$$

Returning to the problem of catalysis, one may express the activation barrier lowering Δ as a sum of components representing *differences* (Δ) of all interaction energy contributions discussed above.

$$\begin{aligned} \Delta &= \Delta E_{\text{TC, C}} - \Delta E_{\text{SC, C}} \\ &= \underbrace{\Delta E_{\text{EL, MTP}}^{(1)} + \Delta E_{\text{EL, PEN}}^{(1)} + \Delta E_{\text{EX}}^{(1)}}_{\Delta E^{(1)}} \\ &\quad + \underbrace{\Delta E_{\text{IND, LE}}^{(R)} + \Delta E_{\text{IND, CT}}^{(R)} + \Delta E_{\text{DISP}}^{(2)}}_{\Delta E^{(R)}} \end{aligned} \quad (11)$$

This way, it is possible to analyse the physical nature of catalytic activity due to any molecular environment interacting with reactants. Next, one may figure out the relative importance of the above discussed (eqn. 11) contributions of Δ and derive simplified methods for studying catalytic activity of much more complex systems. It is obvious that the nature of Δ components (eqn. 11) should not necessarily be the same as for the corresponding interaction energy components $\Delta E_{\text{TC, C}}$ or $\Delta E_{\text{SC, C}}$.

Sample results

To demonstrate the analysis described above, the model reaction $\text{H}_2\text{O} + \text{CO}_2 \rightarrow \text{H}_2\text{CO}_3$ has been selected as a test case, for which the optimal structure of the first solvent shell exerting maximal catalytic activity has been determined in our previous study [15] (Fig. 1).

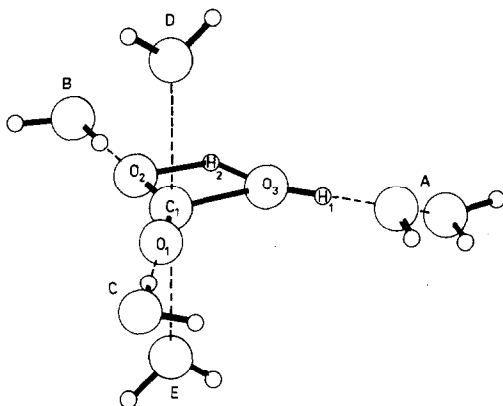


Fig. 1. Optimal locations of solvent molecules (H_2O , OH^-) with respect to transition complex TC (taken from ref. 15).

The best theoretical estimate of the activation barrier for this reaction proceeding in the gas phase is $B_G = 52.2 \text{ kcal mol}^{-1}$ [28], whereas in neutral solution the CO_2 hydration reaction proceeds much faster, with an experimentally determined activation energy $B_C = 17.7 \text{ kcal mol}^{-1}$ [29]. Reaction in alkaline solution has an even lower barrier $B_C = 13.1 \text{ kcal mol}^{-1}$ [30]. Thus the hydration shell evidently exerts some catalytic activity. In our study, the first hydration shell was represented by $C = (\text{H}_2\text{O})_5$ when modeling the neutral solution, or $C = (\text{H}_2\text{O})_4\text{OH}^-$ modeling alkaline environment. Solvent molecules have been positioned at the most stable locations around the transition complex TC (Fig. 1). At such locations they exert highest catalytic activity. Transition complex structure has been taken from [28]. However, recent studies on formaldehyde-water may indicate a nonplanar structure of the transition state [31]. All *ab initio* calculations reported in this study have been performed in split-valence 4-31G basis set [32]. Table 1 presents all Δ contributions defined in the previous section for all constituents of the model solvent shell separately and a two-body estimate of the overall catalytic activity.

The relative importance of various catalytic activity components for $C = (\text{H}_2\text{O})_5$ hydration shell modeling the neutral solution is:

$$|\Delta E_{\text{EL, MTP}}^{(1)}| > |\Delta E_{\text{EX}}^{(1)}| \gg |\Delta E_{\text{DISP}}^{(2)}| > |\Delta E_{\text{IND, CT}}^{(R)}| > |\Delta E_{\text{IND, LE}}^{(R)}| \gg |\Delta E_{\text{EL, PEN}}^{(1)}| \quad (12)$$

TABLE 1

Contributions to activation barrier lowering Δ from all constituents of $(\text{H}_2\text{O})_5$ and $(\text{H}_2\text{O})_4\text{OH}^-$ model solvent shells evaluated within *ab initio* approach (4-31G basis set) (in kcal mol⁻¹)

Solvent	Term	$\Delta E_{\text{EL}, \text{MTP}}^{(1)}$	$\Delta E_{\text{EL}, \text{PEN}}^{(1)}$	$\Delta E_{\text{EX}}^{(1)}$	$\Delta E_{\text{IND}, \text{LE}}^{(R)}$	$\Delta E_{\text{IND}, \text{CT}}^{(R)}$	$\Delta E_{\text{DISP}}^{(2)}$	ΔBSSE	Δ
$\text{H}_2\text{O}(\text{A})$		-13.38	-2.33	9.88	-1.39	-2.91	-2.56	-0.44	-12.69
$\text{OH}^-(\text{A})$		-55.05	-2.73	44.41	-8.72	-19.35	-8.93	-0.63	-50.37
$\text{H}_2\text{O}(\text{B})$		-2.11	0.46	2.60	-0.25	-0.45	-0.84	-0.19	-0.59
$\text{H}_2\text{O}(\text{C})$		-2.52	-0.10	1.27	-0.26	-0.26	-0.26	0.00	-2.12
$\text{H}_2\text{O}(\text{D}, \text{E})$		-2.56	0.88	0.66	-0.12	-0.09	-0.27	-0.25	-1.50
$(\text{H}_2\text{O})_5(\text{A} \cdot \text{E})$		-23.13	-0.20	15.09	-2.15	-3.79	-4.20	-1.14	-18.39
$(\text{H}_2\text{O})_4\text{OH}^-(\text{A} \cdot \text{E})$		-64.80	-0.60	49.62	-9.48	-20.23	-10.57	-1.33	-56.07

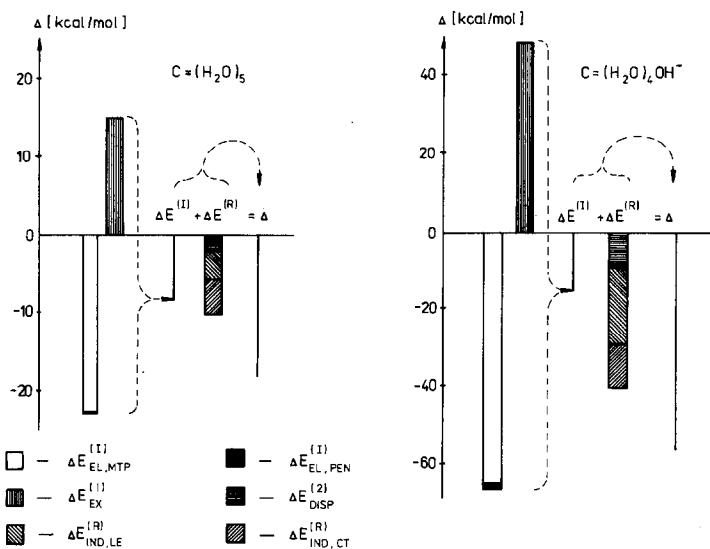


Fig. 2. Components of activation barrier lowering Δ for CO_2 hydration in the presence of solvent C (*ab initio* results in 4-31G basis set).

whereas for $\text{C} = (\text{H}_2\text{O})_4\text{OH}^-$ modeling the alkaline solution we have:

$$|\Delta E_{\text{EL,MTP}}^{(1)}| > |\Delta E_{\text{EX}}^{(1)}| \gg |\Delta E_{\text{IND,CT}}^{(R)}| > |\Delta E_{\text{DISP}}^{(2)}| > |\Delta E_{\text{IND,LE}}^{(R)}| \gg |\Delta E_{\text{EL,PEN}}^{(1)}| \quad (13)$$

As we have already noted [15], the nonphysical *intermolecular* BSSE effect in Δ is relatively small due to cancellation of BSSE in $\Delta E_{\text{TC,C}}$ and $\Delta E_{\text{SC,C}}$, and amounts to *ca.* -1 kcal mol^{-1} . Inclusion of charged OH^- ion in the solvent shell results in enhancement of catalytic activity, in agreement with experimental data [29, 30]. The nature of the catalytic activity for both kinds of hydration shells is illustrated in Fig. 2, where negative Δ constituents represent catalytic and positive inhibitory activity. Although the electrostatic multipole term is undoubtedly the major component of Δ , some other contributions are comparable in magnitude but cancel each other to a great extent. This explains the relative success of earlier studies based on the electrostatic hypothesis [1 - 9]. It is obvious that the more realistic models will require proper representation of the remaining contributions and a more gradual transition from the complete model to more approximate methods. This approach has still all the limitations of the transition state theory (for a recent review see [33, 34]), neglects higher hydration shells and does not take into account the entire dynamics of the solvent shell. Finally, Δ values may depend somewhat on basis set, but certainly not to the degree found for activation barriers [15] and probably less than the observed basis set dependence of interaction energies [11]. Therefore it is not recommended to compare quantitatively the somewhat exaggerated theoretical Δ values with experimental data. In any case, it is clearly evident that the charged constituent of the solvent shell (OH^-) exerts much stronger catalytic

activity than the neutral one (H_2O), and the solvent-transition state complex interaction $\Delta E_{\text{TC},\text{C}}$ is the most effective at site A. Thus the particular arrangement of charged constituents in a molecular environment seems to be a key to understanding the molecular mechanisms of catalytic activity. As has been demonstrated in [13], one may predict the optimal catalyst charge configuration for any reaction from the corresponding difference in the molecular potential map (see also the next section (d)).

Gradually simplified models of catalytic activity

Instead of the very costly variational calculations of activation barriers B_{C} and B_{G} , one may estimate separately the most important additive $\Delta E_{\text{TC},\text{C}}$ and $\Delta E_{\text{SC},\text{C}}$ contributions. In contrast to the variational approach, each term defined within the perturbation method can be evaluated separately from the well-known analytical formulas which can be gradually simplified or approximated.

(a) The most significant and anisotropic electrostatic multipole term $E_{\text{EL},\text{MTP}}^{(1)}$ can be evaluated directly from the multipole expansion. Unfortunately, such expansion, when expressed in terms of *molecular* multipole moments, is not convergent at intermediate distances of chemical interest [35]. This problem may be overcome by dividing the molecular charge distribution into smaller contributions, most naturally atomic segments, and using atomic monopoles q , dipoles μ , quadrupoles Q , etc. derived directly from the *ab initio* wavefunction [12]:

$$\begin{aligned}
 E_{\text{EL},\text{MTP}}^{(1)} = & \sum_{a \in A} \sum_{b \in B} [q_a q_b R_{ab}^{-1} + q_b (\vec{\mu}_a \vec{R}_{ab}) R_{ab}^{-3} - q_a (\vec{\mu}_b \vec{R}_{ab}) R_{ab}^{-3} \\
 & + (\vec{\mu}_a \vec{\mu}_b) R_{ab}^{-3} - 3(\vec{\mu}_a \vec{R}_{ab})(\vec{\mu}_b \vec{R}_{ab}) R_{ab}^{-5} \\
 & + q_a (\vec{R}_{ab} \vec{Q}_a \vec{R}_{ab}) R_{ab}^{-5} + q_b (\vec{R}_{ab} \vec{Q}_b \vec{R}_{ab}) R_{ab}^{-5} + \dots] \quad (14)
 \end{aligned}$$

Terminating the expansion (eqn. 14) at its R_{ab}^{-3} term appears to be sufficient for most chemical applications [2, 24]. It must be noted, however, that the use of the first isotropic monopole-monopole term in eqn. (14) is not enough, because the higher multipoles directly reflect the anisotropy of atomic charge distribution which may sometimes be significant, especially for the atoms with lone electron pairs. In addition, atomic multipoles reflect the unique ground state electronic structure of individual molecules and hardly seem to be transferable. This explains the difficulties encountered in fitting the anisotropic electrostatic component by isotropic potentials [36 - 39], whereas the exchange component is much more transferable [39, 40]. Unfortunately, the analytic expressions for the remaining constituents of interaction energy are much more complex than the expression given above (eqn. 14) for $E_{\text{EL},\text{MTP}}^{(1)}$ and they are quite sensitive to integral approximations [10]. The formulas for short-range terms $E_{\text{EX}}^{(1)}$ and $E_{\text{IND},\text{CT}}^{(2)}$ contain overlap integrals decreasing exponentially with distance. Similar

behavior may also be expected for the electrostatic penetration term $E_{\text{EL, PEN}}^{(1)}$. More extensive discussion of this subject may be found elsewhere [41]. In a separate study [39] we have recently derived analytical atom-atom potentials in the form

$$E = \sum_{a \in A} \sum_{b \in B} (\beta_{ab} + \gamma_{ab} R_{ab}^{-1}) e^{-\delta_{ab} R_{ab}} \quad (15)$$

where atom pair parameters β , γ and δ are fitted to reproduce the non-empirical *ab initio* values of short-range $E_{\text{EL, PEN}}^{(1)}$, $E_{\text{EX}}^{(1)}$ and $E_{\text{IND, CT}}^{(R)}$ components separately. The classical long-range induction component $E_{\text{IND, LE}}^{(R)}$ was represented in the form

$$E_{\text{IND, LE}}^{(R)} = -1/2 \left[\sum_{a \in A} \alpha_a (\vec{E}_a)^2 + \sum_{b \in B} \alpha_b (\vec{E}_b)^2 \right] \quad (16)$$

where $\vec{E}$ denotes the electric field value estimated again within the atomic multipole expansion, and α was fitted to reproduce the nonempirical $E_{\text{IND, LE}}^{(R)}$ values [39].

By representing all main interaction energy contributions by simplified formulas (10), (14), (15) and (16), one may conduct calculations similar to those presented in the previous section for much larger systems at considerably lower cost.

(b) Because $\Delta E_{\text{EL, MTP}}^{(1)}$ and $\Delta E_{\text{EX}}^{(1)}$ constitute the most significant contribution, a more simple first-order model may be constructed:

$$\Delta \approx E_{\text{EL, MTP}}^{(1)} + E_{\text{EX}}^{(1)} \quad (17)$$

This model should be capable of describing properly the accessibility of reactants by solvent or other catalyst molecules, as the exchange contribution is the main repulsive contribution in $\Delta E_{\text{TC, C}}$ or $\Delta E_{\text{SC, C}}$. Thus this approach may be required when the distance between reactants and environment must be described better than in (c).

(c) Δ may be also estimated, at least qualitatively, from $\Delta E_{\text{EL, MTP}}^{(1)}$ only, taking advantage from the previously noted observation that other contributions tend to cancel each other

$$\Delta \approx E_{\text{EL, MTP}}^{(1)} \quad (18)$$

The use of this model should be restricted to ionic or polar reactants and environments where only the relative orientation of reactants with respect to molecular neighbourhood is of interest.

(d) The above electrostatic model (eqn. 18) can be a subject of further approximations. If the entire molecular environment is represented by a set of point charges q_c , we may write

$$\Delta \approx \sum_{c \in C} q_c (V_{\text{TC}} - V_{\text{SC}}) \quad (19)$$

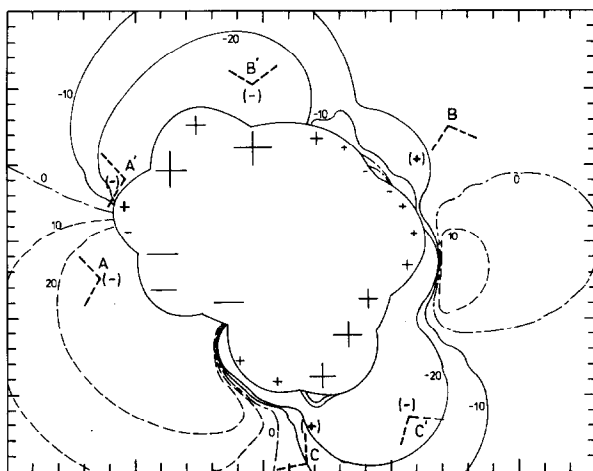


Fig. 3. The difference electrostatic potential map $V_{TC} - V_{SC}$ in the molecular plane of reactants for CO_2 hydration (4-31G basis set).

where V_{TC} , V_{SC} denote the electrostatic molecular potentials of transition state complex TC or substrate complex SC. This representation of Δ , although very crude, was found to be useful in theoretical catalyst design [13]. Namely, the difference of molecular potentials $V_{TC} - V_{SC}$ represents the catalytic activity of a single unitary positive point charge. As has been demonstrated [13], one may also use $V_{TC} - V_{SC}$ to predict catalytic activity for any fractional point charge, including negative charges. Additionally the model may be generalized for multiple point charge representations, as the nonadditive effects are negligible [13]. Thus an optimal charge configuration of molecular environment exerting the maximal catalytic activity may be predicted *a priori* for any chemical reaction (see Figs. 3 and 6 in [13] and Fig. 6 in [42]). Such optimal charge configuration positioned on the van der Waals volume of reactants has been also presented in Fig. 3, where the sizes of + or - signs are proportional to the expected activation energy lowering Δ by the corresponding unit charge (+ or -).

(e) As the prevailing contribution in eqn. (19) arises from frontal (f) solvent atoms engaged in the shortest intermolecular contacts with transition state complex TC, one may write the most crude estimate as:

$$\Delta \cong q_{f \in C}(V_{TC} - V_{SC}) \quad (20)$$

In order to compare the simplified models (a) - (e) with the *ab initio* approach, the corresponding results have been presented in Table 2 for all optimal solvent positions A - E as well as for three other water molecule positions [43] exerting inhibitory activity ($\Delta > 0$) A', B' and C' (Fig. 3). As in ref. 15, the centres of gravity and the orientation of axes of the moments of inertia for TC and SC coincide. One may easily generate other $V_{TC} - V_{SC}$ maps by subtracting properly superimposed V_{TC} (Fig. 4) and V_{SC} (Fig. 5) molecular potential maps. Although Δ estimates in the most complete model a includ-

TABLE 2

Comparison of *ab initio* and simplified model estimates of activation barrier lowerings Δ for first solvent shell constituents positioned at optimal (A, B, C, D and E) or inhibitory (A', B', and C') locations (in kcal mol⁻¹)

Model	<i>ab initio</i> 4-31G	a	b	c	d	e
(Formula)	(4) + (10)	(10) + (15) + (16)	(15) + (16)	(14)	(19) ^b	(20) ^b
Solvent						
H ₂ O (A)	-12.69	-13.46	-5.49	-13.38	-11.00	-24.48
OH ⁻ (A)	-50.37	-59.09	-29.12	-55.05	-49.67	-52.22
H ₂ O (B)	-0.59	-1.12	0.38	-2.11	-2.20	-5.93
H ₂ O (C)	-2.12	-2.23	-1.09	-2.52	-2.81	-8.89
H ₂ O (D, E)	-1.50	-2.45	-2.18	-2.56	-1.87	-5.19
H ₂ O (A')	5.55 ^a	5.96	0.55	6.44	5.26	3.59
H ₂ O (B')	6.60 ^a	8.66	-0.62	7.55	6.08	14.75
H ₂ O (C')	21.24 ^a	22.72	20.63	24.47	20.01	37.67

^a From ref. 43.

^b $V_{TC} - V_{SC}$ estimated from eqn. (21).

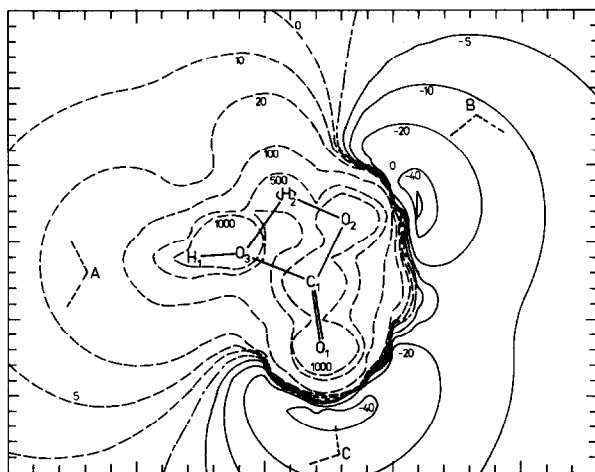


Fig. 4. Electrostatic potential map V_{TC} for transition state complex in its molecular plane (4-31G basis set).

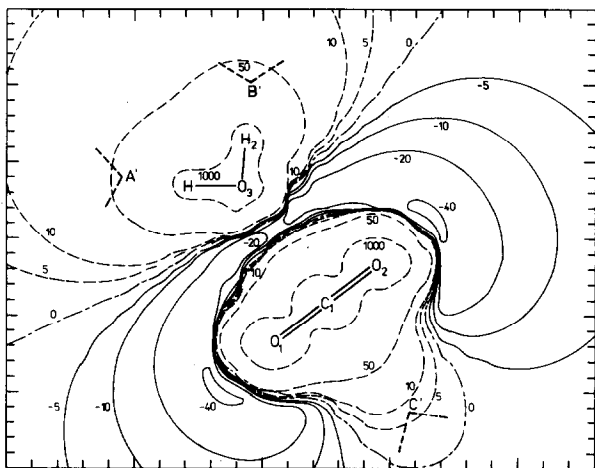


Fig. 5. Electrostatic potential map V_{SC} for substrate complex in its molecular plane (4-31G basis set).

ing all major Δ components are almost consistently underestimated (Δ too negative) they still reflect well the relative magnitude and signs of Δ , which seems to be quite sensitive to solvent position. Model b does not perform so well, and the results in column b indicate the difficulties encountered in gradual reduction of model a into electrostatic model c or d. Surprisingly, the results obtained within electrostatic models c and d are quite close to model a and *ab initio* data. Finally, the most crude and exaggerated (but qualitatively correct) Δ values from model e could be directly estimated from visual inspection of the difference potential map $V_{TC} - V_{SC}$ (Fig. 3). Namely, the Δ sign predicted from the product of the frontal atomic charges q_i (given in parentheses in Fig. 3) and the corresponding difference in

potential values $V_{TC} - V_{SC}$ (estimated from eqn. (21)) agrees qualitatively with more accurate predictions. This qualitative information contained in the difference potential map (Fig. 3) might already be a useful tool in understanding the molecular mechanism of solvent effects, catalytic activity of certain enzymes and design of new catalysts. The somewhat larger error observed in model a for $\text{OH}^-(A)$ ion may originate from the fact that the short-range potentials for the exchange term have been obtained [39] from *ab initio* results for neutral complexes. This may lead to underestimation of repulsive interactions for the much more negative atoms.

The cost of evaluating the electrostatic molecular potentials V_{TC} or V_{SC} may be considerably reduced by employing a cumulative atomic multipole expansion again [12].

$$V_C = \sum_{a \in TC \text{ or } SC} [q_a R_{ac}^{-1} + (\vec{\mu}_a \vec{R}_{ac}) R_{ac}^{-3} + (\vec{R}_{ac} \vec{Q}_a \vec{R}_{ac}) R_{ac}^{-5} + \dots] \quad (21)$$

At this point it must be stressed that terminating the above expansion (eqn. 21) at its first term is not sufficient. Atomic charges q , defined usually within arbitrary population analysis, are strongly basis set dependent — even for extended basis sets with polarization functions [12]. Terminating the expansion (21) at the atomic quadrupole term seems to provide reasonably accurate V estimates outside the van der Waals molecular volume [12].

Thus the difficult computational task to determine the catalytic activity of any molecular environment (including anisotropic) can be reduced to evaluating the cumulative atomic multipole moments for substrates, transition complex and all separate constituents of the catalyst and finding the optimal arrangements of neighboring molecules with respect to TC (or vice versa).

Conclusions

(a) The catalytic activity of a molecular environment represented by activation barrier lowering Δ may be alternatively expressed as the difference of transition complex and substrate complex interaction energies with the catalyst, and decomposed into additive electrostatic, exchange, induction and dispersion components.

(b) Although the first-order electrostatic multipole term $\Delta E_{EL, MTP}^{(1)}$ dominates Δ , other contributions seem to be non-negligible but cancelling each other to a great extent, thus justifying earlier electrostatic hypotheses.

(c) In contrast to the variational approach, all Δ contributions may be evaluated *separately* on different levels of approximation, thereby leading to several gradually simplified models of catalytic activity and opening the possibility to study much larger molecular systems. Model a, representing all significant Δ contributions, seems to be the best among all those tested in this study (a, b, c, d, e).

(d) The qualitative information provided by the models outlined above might provide a useful tool in understanding the detailed molecular mechanism of solvent effects, enzyme catalytic activity and design of biomimetic catalysts. Using the difference molecular potential maps $V_{TC} - V_{SC}$ one may easily find the most catalytically active sites of reactant solvation and the charge distribution of the optimal catalyst for a given reaction. This approach is not limited to solvent effect studies only, and can be extended to other environments (such as enzyme active sites) provided the interactions between reactants and molecular surrounding are not too strong.

Acknowledgements

This work has been sponsored in part by PAN. The author is grateful to Dr. John Blaisdell and Dr. Cary Chabalowski for reading and commenting on the manuscript.

References

- 1 J. O. Noell and K. Morokuma, *Chem. Phys. Lett.*, **36** (1975) 465.
- 2 A. Warshel and M. Levitt, *J. Mol. Biol.*, **103** (1976) 227.
- 3 D. M. Hayes and P. A. Kollman, *J. Am. Chem. Soc.*, **98** (1976) 3335.
- 4 B. M. Rode and R. Fussenegger, *Monatsh. Chem.*, **108** (1977) 703.
- 5 A. Johansson and M. Jäntti, *Finn. Chem. Lett.*, (1976) 46.
- 6 O. Tapia, F. Sussman and E. Poulain, *J. Theor. Biol.*, **71** (1978) 49.
- 7 A. Sawaryn and W. A. Sokalski, *Int. J. Quantum Chem.*, **16** (1979) 293.
- 8 P. T. Van Duijnen, B. T. Thole and W. G. Hol, *J. Biophys. Chem.*, **9** (1979) 273.
- 9 R. P. Sheridan and L. C. Allen, *J. Am. Chem. Soc.*, **103** (1981) 1544.
- 10 W. A. Sokalski and H. Chojnacki, *Int. J. Quantum Chem.*, **13** (1978) 679.
- 11 W. A. Sokalski, P. C. Hariharan and J. J. Kaufman, *J. Phys. Chem.*, **87** (1983) 2803.
- 12 W. A. Sokalski and R. A. Poirier, *Chem. Phys. Lett.*, **98** (1983) 86.
- 13 W. A. Sokalski, *Int. J. Quantum Chem.*, **20** (1981) 231.
- 14 W. A. Sokalski, S. Roszak, P. C. Hariharan and J. J. Kaufman, *Int. J. Quantum Chem.*, **23** (1983) 847.
- 15 W. A. Sokalski, A. Sawaryn and H. Chojnacki, *Int. J. Quantum Chem., Quantum Biol. Symp.*, **10** (1983) 321.
- 16 R. Zahradnik, P. Hobza and J. Sauer, *Proc. Symp. Steric Effects in Biomolecules*, 1981, Eger, Hungary, pp. 327 - 346.
- 17 I. H. Williams, D. Spangler, D. A. Femec, G. M. Maggiora and R. L. Schowen, *J. Am. Chem. Soc.*, **105** (1983) 31.
- 18 B. Jeziorski and W. Kołos, *Int. J. Quantum Chem.*, **12**, Suppl. 1 (1977) 91.
- 19 B. Jeziorski and M. Van Hemert, *Mol. Phys.*, **31** (1976) 713.
- 20 W. Kołos, *Theor. Chim. Acta (Berlin)*, **51** (1979) 219 and references therein.
- 21 S. F. Boys and F. Bernardi, *Mol. Phys.*, **19** (1970) 553.
- 22 K. Kitaura and K. Morokuma, *Int. J. Quantum Chem.*, **10** (1976) 325.
- 23 F. P. Groen and F. B. van Duijneveldt, *Theor. Chim. Acta (Berlin)*, in press.
- 24 W. A. Sokalski, P. C. Hariharan and J. J. Kaufman, *J. Comp. Chem.*, **4** (1983) 506.
- 25 L. Petersson and U. Wahlgren, *Chem. Phys.*, **69** (1982) 185.
- 26 J. Andzelm, M. Kłobukowski and E. Radzio-Andzelm, *J. Comp. Chem.*, **5** (1984) 146.
- 27 S. Roszak, W. A. Sokalski, P. C. Hariharan and J. J. Kaufman, in preparation.

- 28 B. Jönsson, G. Karlström, H. Wennerström, S. Forsen, B. Ross and I. Almlof, *J. Am. Chem. Soc.*, **99** (1977) 4628.
- 29 E. Magid and B. O. Turbeck, *Biochem. Biophys. Acta*, **165** (1968) 515.
- 30 B. R. Pinsent, L. Pearson and F. J. Roughton, *Trans. Faraday Soc.*, **52** (1956) 1512.
- 31 I. H. Williams, D. Spangler, D. A. Femec, G. M. Maggiora and R. L. Schowen, *J. Am. Chem. Soc.*, **102** (1980) 6619.
- 32 R. Ditchfield, W. J. Hehre and J. A. Pople, *J. Chem. Phys.*, **54** (1971) 724.
- 33 K. J. Laidler and M. C. King, *J. Phys. Chem.*, **87** (1983) 2657.
- 34 D. G. Truhlar, W. L. Hase and J. T. Hynes, *J. Phys. Chem.*, **87** (1983) 2664.
- 35 G. Pack, H. Wang and R. Rein, *Chem. Phys. Lett.*, **17** (1972) 381.
- 36 C. Votava and R. Ahlrichs, in B. Pullman (ed.), *Intermolecular Forces*, Reidel, New York, 1981, pp. 133 - 147.
- 37 S. L. Price and A. J. Stone, *Mol. Phys.*, **47** (1982) 1457.
- 38 R. M. Berns and A. van der Avoird, *J. Chem. Phys.*, **72** (1980) 6107.
- 39 W. A. Sokalski, A. Lowrey, S. Roszak, J. Blaisdell, V. Lewchenko, P. C. Hariharan and J. J. Kaufman, in preparation.
- 40 H. J. Böhm and R. Ahlrichs, *J. Chem. Phys.*, **77** (1982) 2028.
- 41 P. Claverie, in B. Pullman (ed.), *Intermolecular Interactions; from Diatomics to Biopolymers*, Wiley, New York, 1978, pp. 69 - 305.
- 42 H. Chojnacki, J. Lipiński and W. A. Sokalski, *Int. J. Quantum Chem.*, **19** (1981) 339.
- 43 W. A. Sokalski and A. Sawaryn, unpublished data.